EDUX / ACADEMIC INTELLIGENCE

FULL EXAM ANALYSIS REPORT

QA Stress Fixture — variable subjects, scores and question counts

All Subjects · FAR / QA-EDGE-2026-01 · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	GENERATED
QA Fixture	QA-0001	2026-11-02	02 Nov 2026

PATTERN	DURATION	SUBJECTS	EXAMINATION STATUS
Mixed format	3 hrs	Physics · Chemistry	Synthetic

TOTAL MARKS OBTAINED

217 / 360 marks

18 correct | 9 incorrect | 3 skipped · Attempted 27 / 30 · Time 90m

SCORE **60.3%**

SUCCESS RATE **66.7%**

ACCURACY **66.7%**

OVERALL ACCURACY **66.7%**

MARKS OBTAINED	MAXIMUM MARKS	CORRECT ANSWERS	INCORRECT ANSWERS
217 of 360	360 Mixed format	18 recorded	9 recorded

SKIPPED QUESTIONS	OVERALL SCORE	ACCURACY	TIME TAKEN
3 unattempted	60.3% percentage	66.7% question intelligence	90m recorded

Difficulty analysis

Questions by level

Easy	10 Q
Medium	12 Q
Hard	8 Q

Easy 82% · Medium 66% · Hard 41%

Mistake distribution

Error categories

Concept Error	5
Calculation	4
Formula Error	3
Time Pressure	2

Counts reproduced from the attempt payload.

Subject-wise performance

Accuracy by subject

Physics	67%
Chemistry	53%

Physics 121 / 180 · Chemistry 96 / 180

PERFORMANCE BRIEF / EDITORIAL SUMMARY

This is a synthetic edge-case fixture used only to validate pagination, mathematical typesetting, long labels, missing optional values and dynamic subject combinations. It is not a student report.

REPORT MAP / CONTENTS

01 **Performance breakdown**
30 questions · 4 error categories · 3 difficulty bands

02 **Subject & chapter intelligence**
2 subjects · 8 chapters · mastery classifications

03 **Time, comparison and projection**
90m recorded · 2 timing sections

04 **Question-by-question analysis**
30 review rows · full source explanations and answer comparison

05 **Mistake pattern and evidence**
4 categories · 2 source evidence notes

06 **Revision strategy**
2 subject groups · 2 published priorities

01 ANALYTICS

Performance breakdown

Response composition, difficulty, error patterns and subject marks — all values are reproduced from the examination payload.

Correct versus incorrect

30 questions in the paper



- Correct 18 18
- Incorrect 9 9
- Skipped 3 3

Mistake categories

Count by classification

Concept Error	5
Calculation Error	4
Formula Error	3
Time Pressure	2

Difficulty distribution

Questions, attempted counts, accuracy and average time by difficulty level.

LEVEL	QUESTIONS	ATTEMPTED	ACCURACY	AVG. TIME
Easy	10	10	82%	2.1 min
Medium	12	11	66%	3.4 min
Hard	8	6	41%	5.2 min

Subject-wise marks and accuracy

Scores, pace, attempt quality and evidence areas.

Subject performance — marks, accuracy, pace and evidence areas

SUBJECT	MARKS	ACCURACY	TIME	ATT.	CORR.	RANK	LEVEL	WEAK AREAS	STRONG AREAS
Physics	121 / 180	67%	48	18	12	2	Medium	Rotational Motion	Kinematics
Chemistry	96 / 180	53%	42	9	6	4	Hard	Chemical Kinetics	Mole Concept

Performance signals

Additional headline metrics published with this attempt.

GRADE B	RANK 91	PERCENTILE 88.2	BATCH RANK 4	COHORT SIZE 60
OVERALL SPEED 2.8	CONFIDENCE INDEX 74	READINESS SCORE 70	HEALTH SCORE 79	NEGATIVE MARKS 9
GUESS ATTEMPTS 2	ATTEMPT RATIO 90%	TOTAL QUESTIONS 30	SUCCESS RATE 66.7%	BADGE Synthetic QA

02 MASTERY MAP

Subject and chapter intelligence

Chapter-level accuracy and mastery are ordered exactly as supplied by the analysis engine.

Subject performance — marks, accuracy, pace and evidence areas

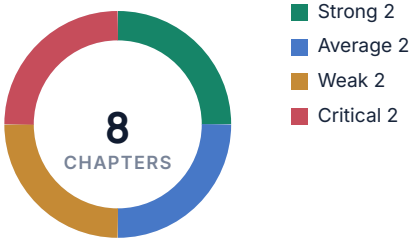
SUBJECT	MARKS	ACCURACY	TIME	ATT.	CORR.	RANK	LEVEL	WEAK AREAS	STRONG AREAS
Physics	121 / 180	67%	48	18	12	2	Medium	Rotational Motion	Kinematics
Chemistry	96 / 180	53%	42	9	6	4	Hard	Chemical Kinetics	Mole Concept

Chapter mastery — accuracy, marks and time

CHAPTER	SUBJECT	ACCURACY	MARKS	TIME	ATT. %	MASTERY
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	42%	4	3m	70	Critical
Chemical Kinetics 2	Chemistry	48%	5	4m	73	Weak
Rotational Motion 3	Physics	54%	6	5m	76	Average
Chemical Kinetics 4	Chemistry	60%	7	6m	79	Strong
Rotational Motion 5	Physics	66%	8	7m	82	Critical
Chemical Kinetics 6	Chemistry	72%	9	8m	85	Weak
Rotational Motion 7	Physics	78%	10	9m	88	Average
Chemical Kinetics 8	Chemistry	84%	11	10m	91	Strong

Mastery distribution

8 chapters classified



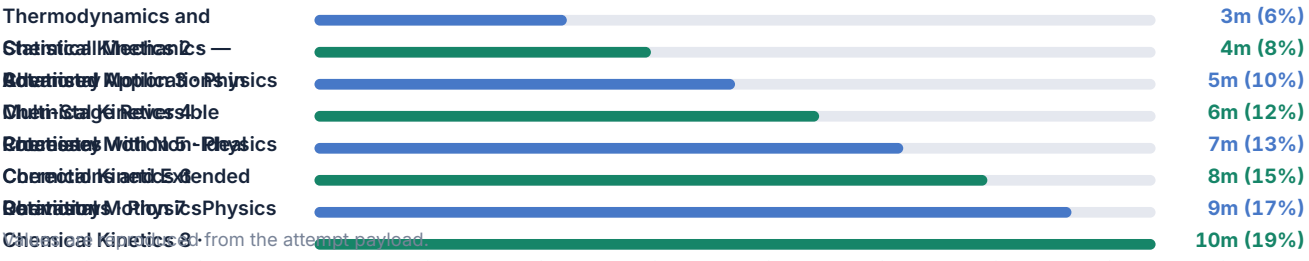
03 PACE & TRAJECTORY

Time, comparison and projection

Timing and comparative signals are shown as published; no trend or score has been recomputed.

Chapter time consumption

Recorded time and share of total chapter time



Chapter names not available from the attempt payload.

Time intelligence

Fastest: Q1 — Kinematics · 0.7 min | Slowest: Q29 — Definite Integration · 7.9 min

SECTION	ALLOCATED (MIN)	USED (MIN)	EFFICIENCY
Physics	60	48	125
Chemistry	60	42	143

AVG. TIME / QUESTION 3	NAVIGATION COUNT 11	TIME MANAGEMENT 68	FASTEST QUESTION Q1 — Kinematics
FASTEST TIME 0.7	SLOWEST QUESTION Q29 — Definite Integration	SLOWEST TIME 7.9	

Comparative performance

Cohort and historical benchmarks attached to this attempt.

BENCHMARK	SCORE	PERCENTILE	GRADE	REFERENCE
Previous test	198	84.1	B	—
Batch average	176	80.2	—	—

DELTA	VALUE	UP FLAG
vs previous test	+19	true

Projection and readiness

Expected improvement +3.1% · risk Medium

EXAMINATION / PROJECTION	SCORE
QA 1	190
QA 2	217

JEE PERCENTILE 91.1	NEET SCORE —	EXPECTED AIR 21400
TARGET PROBABILITY 77	RISK LEVEL Medium	EXPECTED IMPROVEMENT +3.1%

04 QUESTIONINTELLIGENCE

Question-by-question analysis

30 question-review rows supplied in the source payload. Missing optional fields are shown as — and are never inferred.

Correct | Incorrect | Skipped

Each card preserves question metadata, answer comparison, marks, time, mistake category and the full source explanation.

Q1 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Non-equilibrium reversible heat transfer with variable coefficients	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 1: $x^1 + \sqrt{1} + \sum_{k=1}^{1} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
—	—	4 · —

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q2 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 2: $x^2 + \sqrt{2} + \sum_{k=1}^{2} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	-1 · 1m 54s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Q3 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 3: $x^{3} + \sqrt{3} + \sum_{k=1}^{3} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
C	D	4 · 2m 36s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_{0}$.

Q4 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 4: $x^{4} + \sqrt{4} + \sum_{k=1}^{4} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
D	A	4 · 3m 18s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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\frac{1}{2
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is malformed and must be shown verbatim for verification.

Q5 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 5: $x^5 + \sqrt{5} + \sum_{k=1}^{5} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
A	B	-1 · 4m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

Q6 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 6: $x^6 + \sqrt{6} + \sum_{k=1}^{6} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	4 · 4m 42s

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Q7 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 7: $x^{7} + \sqrt{7} + \sum_{k=1}^{7} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
C	D	4 · 5m 24s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q8 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 8: $x^{8} + \sqrt{8} + \sum_{k=1}^{8} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
—	—	-1 · —

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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\frac{1}{2
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is malformed and must be shown verbatim for verification.

Q9

Q9 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 9: $x^{9} + \sqrt{9} + \sum_{k=1}^{9} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
A	B	4 · 6m 48s

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

Q10

Q10 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 10: $x^{10} + \sqrt{10} + \sum_{k=1}^{10} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	4 · 1m 12s

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q11 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 11: $x^{11} + \sqrt{11} + \sum_{k=1}^{11} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
C	D	-1 · 1m 54s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$.

Q12 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 12: $x^{12} + \sqrt{12} + \sum_{k=1}^{12} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
D	A	4 · 2m 36s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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is malformed and must be shown verbatim for verification.

Q13 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 13: $x^{13} + \sqrt{13} + \sum_{k=1}^{13} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
A	B	4 · 3m 18s

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q14 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 14: $x^{14} + \sqrt{14} + \sum_{k=1}^{14} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	-1 · 4m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Q15 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 15: $x^{15} + \sqrt{15} + \sum_{k=1}^{15} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
—	—	4 · —

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$.

Q16 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 16: $x^{16} + \sqrt{16} + \sum_{k=1}^{16} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
D	A	4 · 5m 24s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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is malformed and must be shown verbatim for verification.

Q17 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 17: $x^{17} + \sqrt{17} + \sum_{k=1}^{17} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
A	B	-1 · 6m 06s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

Q18 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 18: $x^{18} + \sqrt{18} + \sum_{k=1}^{18} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	4 · 6m 48s

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Q19 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 19: $x^{19} + \sqrt{19} + \sum_{k=1}^{19} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
C	D	4 · 1m 12s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q20 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 20: $x^{20} + \sqrt{20} + \sum_{k=1}^{20} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
D	A	-1 · 1m 54s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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\frac{1}{2
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is malformed and must be shown verbatim for verification.

Q21 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Non-equilibrium reversible heat transfer with variable coefficients	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 21: $x^{21} + \sqrt{21} + \sum_{k=1}^{21} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
A	B	4 · 2m 36s

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

Q22 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 22: $x^{22} + \sqrt{22} + \sum_{k=1}^{22} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
—	—	4 · —

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Q23

Q23 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 23: $x^{23} + \sqrt{23} + \sum_{k=1}^{23} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
C	D	-1 · 4m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$.

Q24

Q24 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 24: $x^{24} + \sqrt{24} + \sum_{k=1}^{24} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
D	A	4 · 4m 42s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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\frac{1}{2
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is malformed and must be shown verbatim for verification.

Q25 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 25: $x^{25} + \sqrt{25} + \sum_{k=1}^{25} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
A	B	4 · 5m 24s

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q26 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 26: $x^{26} + \sqrt{26} + \sum_{k=1}^{26} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	-1 · 6m 06s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Q27 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 27: $x^{27} + \sqrt{27} + \sum_{k=1}^{27} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
C	D	4 · 6m 48s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$.

Q28 · Correct

Correct

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Chemistry	Chemical Kinetics	Projectile motion	Easy	MCQ

QUESTION / OPTIONS

Question: Evaluate the expression for case 28: $x^{28} + \sqrt{28} + \sum_{k=1}^{28} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
D	A	4 · 1m 12s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

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\frac{1}{2
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is malformed and must be shown verbatim for verification.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q29 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients	Medium	Numerical

QUESTION / OPTIONS

Question: Evaluate the expression for case 29: $x^{29} + \sqrt{29} + \sum_{k=1}^{29} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
—	—	-1 · —

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.

Q30 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC	DIFFICULTY	TYPE
Physics	Rotational Motion	Projectile motion	Hard	Assertion Reason

QUESTION / OPTIONS

Question: Evaluate the expression for case 30: $x^{30} + \sqrt{30} + \sum_{k=1}^{30} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS · TIME
B	C	4 · 2m 36s

SOLUTION / EXPLANATION

EXPLANATION

Using the standard summation identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved.

Question data completeness

30 questions are recorded in the paper intelligence summary; 30 per-question review rows were supplied for publication.

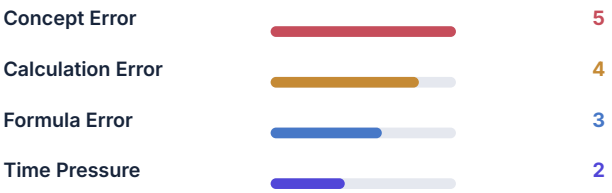
05 ERRORINTELLIGENCE

Mistake pattern and evidence

Mistake categories, the question-level evidence attached to them, and topic-level accuracy.

Mistake pattern

Count by category



Response composition

30 questions



Question-level mistake evidence

Details are reproduced verbatim from the source payload.

Q	SUBJECT	CATEGORY	TOPIC	SOURCE DETAIL
Q4	Physics	Concept Error	Definite Integration	Misapplied substitution — used u = x^2 instead of u = x^2+1
Q11	Chemistry	Calculation Error	Chemical Kinetics	Decimal error in $1/r^2$ computation

Topic-level accuracy

6 topics with published mastery levels.

TOPIC	SUBJECT	ACCURACY	LEVEL
Definite Integration	Physics	44%	Critical
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	51%	Weak
Bayes theorem	Physics	62%	Average
Coordination compounds	Chemistry	48%	Weak
Chemical Kinetics	Chemistry	82%	Strong
Mole Concept	Chemistry	91%	Strong

06 LEARNINGINTELLIGENCE

Revision strategy

Revision priorities are ordered weakest first. Critical topics are highlighted with a fine rule, not an aggressive warning block.

01 Physics

67% subject accuracy · Review

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Definite Integration	—	44%	Critical
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	—	51%	Weak
Bayes theorem	—	62%	Review

02 Chemistry

53% subject accuracy · Weak

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Coordination compounds	—	48%	Weak

Published revision priorities

Priority, topic and timeframe from the recommendation engine.

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Definite Integration — substitution	Critical	48 hours
Chemical Kinetics — rate laws	High	5 days

Subject-wise weak topics and chapters

Names are reproduced exactly as supplied by the recommendation engine.

Weak chapters

NO.	SOURCE VALUE
1	Calculus (44%)
2	Chemical Kinetics (48%)

Weak topics

NO.	SOURCE VALUE
1	Definite Integration
2	Coordination compounds
3	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations

Recommended next actions

Practice, PYQ, mock and lecture resources attached to this attempt.

ACTION	RESOURCE	DETAIL
PYQ	JEE Main 2024 — Integration (5 Qs)	PYQ Bank
Practice	20-question drill — substitution	20 · Medium

ACTION	RESOURCE	DETAIL
Mock test	Sectional mock — Chemistry	Sectional
Lecture	Kinetics masterclass	Video

07 DATAINTEGRITY

Source data register

49 unique source values are listed below with their payload paths. This register guarantees that no field was lost in the redesign.

SOURCE FIELD	SOURCE VALUE
meta.course Synthetic validation paper	attempt.elapsedSeconds 5400
questionReview[0].text Evaluate the expression for case 1: $x^{1} + \sqrt{1} + \sum_{{k=1}}^{1} k$.	questionReview[0].explanation The integral reduces directly: $$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$ The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$.
questionReview[0].hint Check the sign convention and the limits of integration carefully.	questionReview[1].time 1.9
questionReview[1].text Evaluate the expression for case 2: $x^{2} + \sqrt{2} + \sum_{{k=1}}^{2} k$.	questionReview[1].options[0] 2 s
questionReview[1].options[1] √3 s	questionReview[1].options[2] 4 s
questionReview[1].options[3] A very long alternative answer that must wrap without clipping the card	questionReview[1].explanation Using the standard summation identity $$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$ and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\ce{H2SO4}$ and 1.6×10^{-19} C is preserved.
questionReview[2].time 2.6	questionReview[2].text Evaluate the expression for case 3: $x^{3} + \sqrt{3} + \sum_{{k=1}}^{3} k$.
questionReview[2].explanation A multi-line derivation: $$\begin{aligned} a &= b + c \\ d &= e + f \\ \therefore\; a+d &= b+c+e+f \end{aligned}$$ with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$.	questionReview[3].time 3.3
questionReview[3].text Evaluate the expression for case 4: $x^{4} + \sqrt{4} + \sum_{{k=1}}^{4} k$.	questionReview[3].explanation FLAGGED SOURCE — do not correct: $\frac{1}{2$ is malformed and must be shown verbatim for verification.
questionReview[4].text Evaluate the expression for case 5: $x^{5} + \sqrt{5} + \sum_{{k=1}}^{5} k$.	questionReview[5].time 4.7
questionReview[5].text Evaluate the expression for case 6: $x^{6} + \sqrt{6} + \sum_{{k=1}}^{6} k$.	questionReview[6].time 5.4
questionReview[6].text Evaluate the expression for case 7: $x^{7} + \sqrt{7} + \sum_{{k=1}}^{7} k$.	questionReview[7].text Evaluate the expression for case 8: $x^{8} + \sqrt{8} + \sum_{{k=1}}^{8} k$.
questionReview[8].time 6.8	questionReview[8].text Evaluate the expression for case 9: $x^{9} + \sqrt{9} + \sum_{{k=1}}^{9} k$.
questionReview[9].time 1.2	questionReview[9].text Evaluate the expression for case 10: $x^{10} + \sqrt{10} + \sum_{{k=1}}^{10} k$.
questionReview[10].text Evaluate the expression for case 11: $x^{11} + \sqrt{11} + \sum_{{k=1}}^{11} k$.	questionReview[11].text Evaluate the expression for case 12: $x^{12} + \sqrt{12} + \sum_{{k=1}}^{12} k$.
questionReview[12].text Evaluate the expression for case 13: $x^{13} + \sqrt{13} + \sum_{{k=1}}^{13} k$.	questionReview[13].text Evaluate the expression for case 14: $x^{14} + \sqrt{14} + \sum_{{k=1}}^{14} k$.
questionReview[14].text Evaluate the expression for case 15: $x^{15} + \sqrt{15} + \sum_{{k=1}}^{15} k$.	questionReview[15].text Evaluate the expression for case 16: $x^{16} + \sqrt{16} + \sum_{{k=1}}^{16} k$.

SOURCE FIELD	SOURCE VALUE
questionReview[16].time 6.1	questionReview[16].text Evaluate the expression for case 17: $x^{17} + \sqrt{17} + \sum_{{k=1}}^{17} k$.
questionReview[17].text Evaluate the expression for case 18: $x^{18} + \sqrt{18} + \sum_{{k=1}}^{18} k$.	questionReview[18].text Evaluate the expression for case 19: $x^{19} + \sqrt{19} + \sum_{{k=1}}^{19} k$.
questionReview[19].text Evaluate the expression for case 20: $x^{20} + \sqrt{20} + \sum_{{k=1}}^{20} k$.	questionReview[20].text Evaluate the expression for case 21: $x^{21} + \sqrt{21} + \sum_{{k=1}}^{21} k$.
questionReview[21].text Evaluate the expression for case 22: $x^{22} + \sqrt{22} + \sum_{{k=1}}^{22} k$.	questionReview[22].text Evaluate the expression for case 23: $x^{23} + \sqrt{23} + \sum_{{k=1}}^{23} k$.
questionReview[23].text Evaluate the expression for case 24: $x^{24} + \sqrt{24} + \sum_{{k=1}}^{24} k$.	questionReview[24].text Evaluate the expression for case 25: $x^{25} + \sqrt{25} + \sum_{{k=1}}^{25} k$.
questionReview[25].text Evaluate the expression for case 26: $x^{26} + \sqrt{26} + \sum_{{k=1}}^{26} k$.	questionReview[26].text Evaluate the expression for case 27: $x^{27} + \sqrt{27} + \sum_{{k=1}}^{27} k$.
questionReview[27].text Evaluate the expression for case 28: $x^{28} + \sqrt{28} + \sum_{{k=1}}^{28} k$.	questionReview[28].text Evaluate the expression for case 29: $x^{29} + \sqrt{29} + \sum_{{k=1}}^{29} k$.
questionReview[29].text Evaluate the expression for case 30: $x^{30} + \sqrt{30} + \sum_{{k=1}}^{30} k$.	

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End of report · FAR / QA-EDGE-2026-01

Academic data reproduced without inference. Missing values are shown as —.